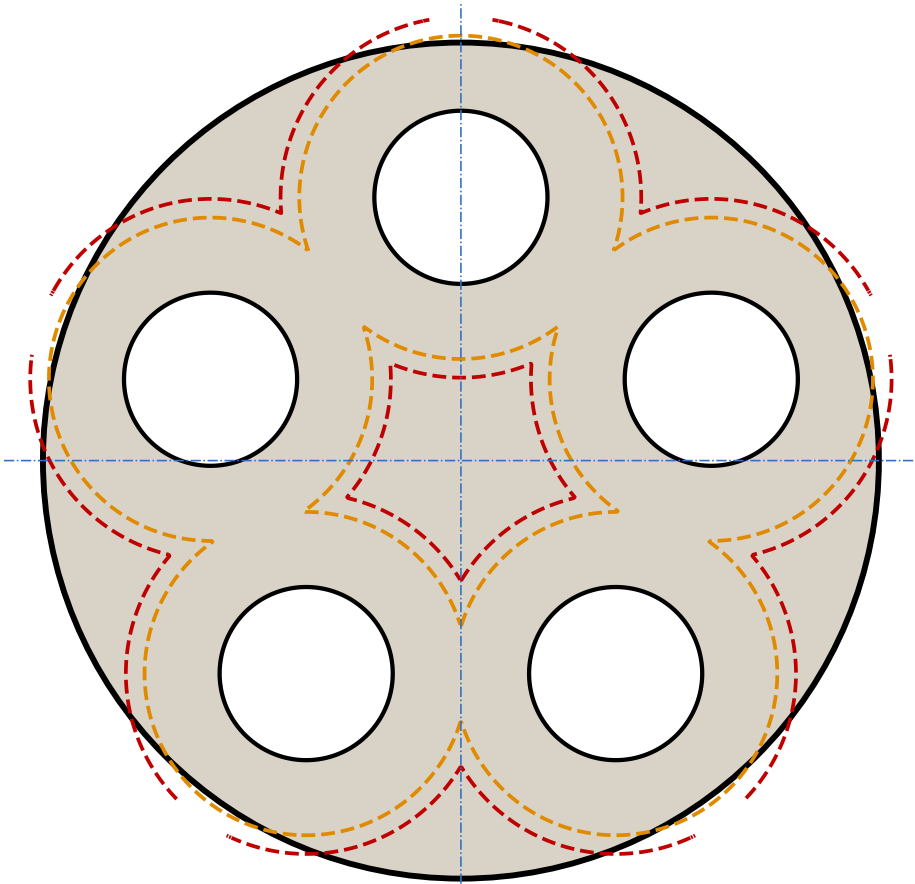


V2.2 FUEL GRAIN - PORT LAYOUT OPTIONS (cross-section, mm)

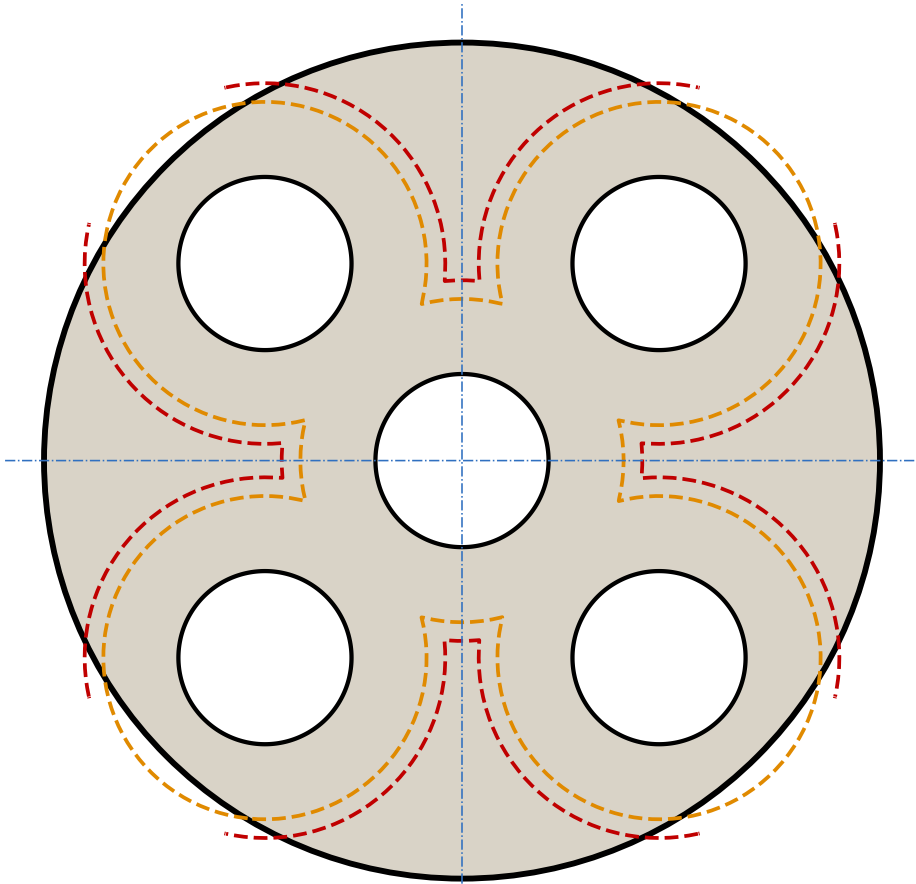
ø85.85 OD x 177.8 long paraffin. All three have the same total port area (1241 mm²), so the same initial oxidiser flux and fuel load.

1 FIVE ROUND PORTS ON A CIRCLE



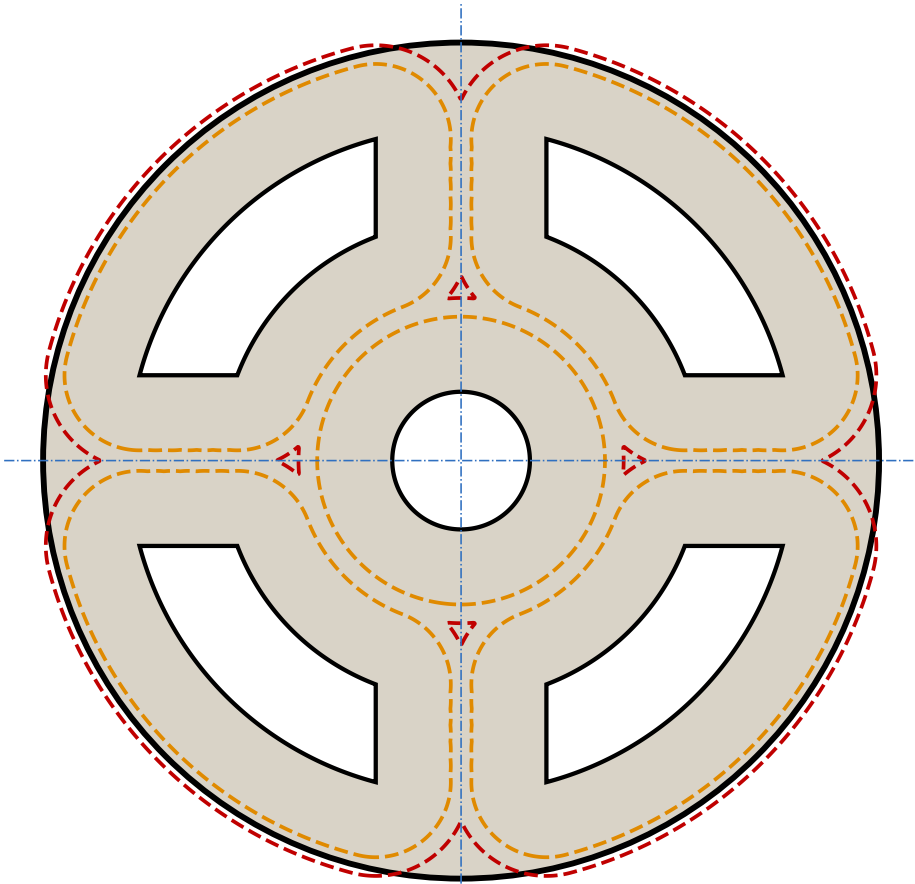
5 x ø17.78 on R27.04 (72° pitch)

2 CENTRE + FOUR ROUND PORTS



5 x ø17.78; ring on R28.62 (90° pitch)

3 CENTRE PORT + FOUR SECTOR PORTS



centre ø14.14; sectors R24.60 to R34.16
spokes and ring web 17.53 thick

layout	port area mm²	perimeter mm	burn surface cm²	wall web mm	ports merge after mm	SP7 a to liquid-out 7.7 mm	SP7 a full burn 9.6 mm
1	1241	279	497	7.00	7.00	THROUGH	THROUGH
2	1241	279	497	5.42	5.42	THROUGH	THROUGH
3	1241	351	625	8.76	8.76	OK	THROUGH

Dashed lines: port surface after the regression the SP7 a (likely N2O) fit gives - orange at liquid depletion (7.7 mm), red at end of the full burn (9.6 mm). Where a dashed line crosses the OD, the grain burns through to the case. Regression distances come from HASTE's circular-port model; layout 3 matches port area (so G_{ox}) and hydraulic diameter, so per-surface regression is comparable. Sector corners are drawn sharp - cast them with ~2 mm fillets. Not a manufacturing drawing.